

Jay Niketan Pathare

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EXPERIENCE

Graduate Student Developer (CSE 611 Project) | Team of 5 | HeinOnline, Buffalo

January 2026 – April 2026

- Engineered a fault-tolerant OCR/LLM metadata extraction pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across three independent sources: CrossRef API, web scrapers, and OCR/LLM vision to extract article titles, authors, DOIs, page ranges, and ToC data from 47 law journals (28,167+ pages).
- Built a fuzzy-matching combinator engine to merge asynchronous JSON outputs into a validated YAML structure, cutting manual data-entry workload by ~90% and raising extraction accuracy from 57.03% to 90.81% over five iterations (43 of 47 journals $\geq 80\%$, 8 at 100%; benchmarked against ground truth).
- Parallelized the pipeline via a worker pool (cores – 1) that pulls the next journal from a shared queue on completion, and added an “aiscore” system to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Led delivery of platform modules end to end: gathered requirements in client meetings, prototyped to validate scope, decomposed the finalized work across a small team, and reviewed and merged their contributions into production.
- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records, and designed a real-time WebSocket + Redis pipeline feeding IoT baby-warmer sensor data into PostgreSQL with ~2s live monitoring.
- Delivered a telemedicine consultation feature (WebRTC) connecting on-site nurses with remote doctors to review test results and procedures with patients, alongside secure PDF report generation.
- Integrated a color-blindness diagnostic module (Ishihara plate testing, ~97–99% clinical sensitivity) with automated reporting and real-time result visualization for practitioners.
- Architected relational schemas and SQL queries for efficient data retrieval, updates, and reporting.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India

November 2023 – May 2024

- Designed and deployed scalable AWS infrastructure (EC2, RDS) with VPC networking and security groups, ensuring secure, low-latency communication for backend services.
- Automated cloud resource provisioning, reducing setup time by ~40% and minimizing deployment-related issues.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Developed an ETL workflow using Python (Pandas, NumPy) and MySQL to process and clean large business datasets, generating reports highlighting key operational trends.

EDUCATION

Master’s in Computer Science

August 2025 – February 2027

University at Buffalo, The State University of New York, Buffalo, NY

GPA: 3.85 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society’s Institute of Technology, Mumbai University, India

August 2019 – May 2023

TECHNICAL SKILLS

- Languages & Databases:** Python, C, SQL, PostgreSQL, MySQL, HTML/CSS
- Frameworks:** Django, Django REST Framework, Django Channels, ReactJS, PyTorch, scikit-learn, Pandas, NumPy
- AI/ML:** LLM Integration, Prompt Engineering, Transformers, Machine Learning, Deep Learning, Agentic AI, Model Evaluation, GraphSAGE, GAT, Neural Networks (CNN, GNN)
- Software Engineering:** OOP, REST APIs, Full-Stack Development, Backend Development, Concurrency, System Design, ETL, Load Testing, Backtracking Algorithms, Agile, Parallel Processing, Real-Time Monitoring
- Tools & Cloud:** Linux, Docker, Redis, AWS (EC2/RDS), Git, GitHub, GitHub Copilot, Locust, OAuth/Auth0, WebRTC, WebSockets, Gemini API, Groq API, Tableau, Power BI

ACADEMIC PROJECTS

LLM Application Optimization Engine | Individual Project

September 2026

- Built a multi-model pipeline generating tailored resumes, cover letters, and independent fit assessments, using exact dynamic programming to select 3 of 6 projects across 60 valid allocations.
- Reduced end-to-end runtime by ~73% (11 to ~3 minutes) through three concurrent LLM workers, account rotation, and bounded retries, with deterministic checks rejecting unsupported claims.
- Developed a 2,500+ test pytest suite with invariant and metamorphic tests for extraction, grounding, and optimization; verified generated PDFs by compiling LaTeX and re-extracting rendered text.

Decoupled Learning Management System (LMS) | Individual Project

July 2026

- Architected a Django REST Framework and React LMS with Redis-backed Django Channels for live quizzes, leaderboards, and rate-limited chat, supporting 45+ concurrent users with sub-500ms latency.
- Secured REST and WebSocket access with stateless JWT, implemented idempotent quiz grading, and load-tested with Locust and asyncio, reducing p95 REST latency by 26% and quiz-submission time from 2000ms to 1500ms.

Fraud Detection using Graph Neural Networks | Individual Project | University at Buffalo, NY

March 2026

- Trained a GraphSAGE-based fraud detection model on the IEEE-CIS dataset (590k+ records), using Focal Loss to handle a severe 27:1 class imbalance and achieving a 0.9259 AUC-ROC (vs. 0.8584 MLP baseline, 0.7928 GAT).
- Converted tabular transaction data into a graph network using shared card, device, and email identifiers, capping node groups to prevent memory explosion while capturing relational fraud rings.

CERTIFICATIONS, AWARDS & PUBLICATIONS

- AWS Cloud Practitioner** July 2024
- Certificate of Appreciation:** CSE Workshop “GitHub: Hands-On from Basics to Advanced,” UB February 2026
- Research Publication:** J. Pathare et al., “Music Genre Classification,” IJRAR, Apr. 2023 (97.68% accuracy). Link.